

A Predictive Data Quality Scoring Model for Customer Onboarding Data in Namibian Commercial Banks

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Chapter 1 - INTRODUCTION

1.1 Background

Customer onboarding in the Namibian banking sector is a critical process involving the collection and validation of sensitive customer data, such as national identity documents, proof of address, and income details, to comply with Know Your Customer (KYC) and Anti-Money Laundering (AML) regulations enforced by the Bank of Namibia (Financial Intelligence Act, 2012; Bank of Namibia, 2024). Despite ongoing digital transformation within the banking industry, banks continue to process large volumes of onboarding applications annually while facing operational challenges caused by data quality issues such as incomplete forms, invalid identity formats, and inconsistent information.

These inefficiencies stem primarily from the current hybrid onboarding model, which combines digital self-service platforms and assisted in-branch data entry. In both cases, data validation remains largely reactive, with errors typically identified only after submission into core banking systems (Redman, 2022; Nshimba & Lutu, 2023). This creates a detection lag that delays verification processes while increasing operational and compliance risks.

With the growing adoption of digital banking services, ensuring high-quality data at the point of entry has become increasingly important. Machine learning provides an opportunity to shift from reactive validation approaches to proactive data quality management by predicting and flagging potential errors in real time (Kaur & Singh, 2021). This aligns with national initiatives such as the Bank of Namibia Strategic Plan (2024–2026) and Namibia's Vision 2030, which promote secure digital financial services and electronic Know Your Customer (e-KYC) systems.

This study aims to develop a predictive data quality scoring model for customer onboarding data in the banking sector. The proposed model will use supervised machine learning techniques to assess data quality across key dimensions, including completeness, accuracy, consistency, timeliness, validity and uniqueness. These dimensions are critical in banking environments where inaccurate customer information may negatively affect compliance, fraud detection, and service delivery. The model will assign reliability scores to data at the point of entry, helping reduce

manual verification efforts, improve onboarding efficiency, and strengthen regulatory compliance within the Namibian banking context.

1.2 Problem Statement

In an ideal banking environment, customer data is accurate and verified at the point of entry to ensure regulatory compliance and operational efficiency. However, this is not always the case within the Namibian commercial banking sector, where data quality is still largely managed through reactive validation approaches. As a result, banks continue to experience data quality failures during onboarding, where incomplete, inconsistent, or erroneous customer information contributes to processing delays, operational inefficiencies, and increased risk exposure (Redman, 2022; Nshimba & Lutu, 2023).

These errors, whether originating from digital self-service applications or manual in-branch data entry, are often only identified during back-office audits or after transaction processing failures occur. This creates a detection lag that delays verification processes while increasing operational costs and the risk of non-compliance with the Financial Intelligence Act (2012). In addition, poor onboarding experiences caused by repeated corrections and delays may contribute to customer dissatisfaction and onboarding abandonment (Validata, 2023).

Although banks currently use rule-based validation systems, these approaches remain limited in detecting human-related errors that bypass standard checks, such as logical inconsistencies, unusual behavioural patterns, duplicate entries, or invalid input formats. Traditional validation approaches are primarily deterministic and therefore struggle to identify behavioural irregularities and context-based data inconsistencies in real time. Furthermore, manual validation processes are not scalable within a rapidly growing digital banking environment where onboarding volumes continue to increase (Kaur & Singh, 2021).

Currently, there is limited research and no widely adopted proactive mechanism within the Namibian banking sector that leverages historical error patterns and real-time behavioural inputs to predict onboarding data quality before information is

permanently recorded within banking systems. Without such predictive capabilities, banks may continue experiencing increasing operational costs, onboarding inefficiencies, and compliance challenges. This highlights the need for a proactive and intelligent data quality assessment approach capable of detecting onboarding errors at the point of data entry.

1.3 Research Objectives

1.3.1 Main Objective

To develop a predictive data quality scoring model that enhances onboarding data integrity at the point of ingestion within Namibian commercial banks.

1.3.2 Sub-objectives

1. To identify and prioritise key data quality dimensions relevant to the Namibian banking onboarding environment, including completeness, accuracy, consistency, timeliness, validity, and uniqueness.
2. To analyse and categorise common human-centric data entry errors occurring across digital self-service and staff-assisted retail onboarding channels.
3. To develop a machine learning-based scoring model capable of assigning a real-time reliability score to onboarding data at the point of ingestion.
4. To evaluate the effectiveness of the proposed model in reducing data errors, improving early error detection, and enhancing onboarding efficiency compared to traditional validation approaches.

1.4 Research questions

1. How can predictive modelling improve data quality at the point of entry in Namibian commercial banks?
2. What data quality dimensions are most critical to ensuring onboarding data integrity in Namibian commercial banks?
3. How can behavioural indicators such as typing speed, dwell time, and input latency be used to detect data entry errors in real time?
4. To what extent can a machine learning-based model reduce data errors and detection lag compared to traditional validation methods?

5. How effective is a predictive data quality scoring model in improving onboarding efficiency and data integrity in Namibian banks?

1.4 Motivation for the Study

The motivation for this research stems from the need to safeguard the integrity of the Namibian financial system as it transitions toward a digital and multi-channel customer onboarding environment. The rapid growth of fintech and mobile banking services has increased onboarding volumes while exposing limitations in traditional data validation and processing mechanisms (Bank of Namibia, 2023; World Bank, 2022).

Currently, many banking systems rely on reactive validation approaches, where data errors are detected only after submission or during back-office processing. This results in operational inefficiencies, increased processing costs, and delays in identifying and correcting inaccurate customer information (Redman, 2022; Kaur & Singh, 2021). A shift toward proactive data quality management has the potential to reduce reliance on manual correction processes and improve overall operational efficiency.

These challenges also raise concerns regarding regulatory compliance under the Financial Intelligence Act (2012), as inaccurate or incomplete customer data may increase exposure to financial crime risks and regulatory penalties (Financial Intelligence Act, 2012). Furthermore, delayed detection of onboarding errors negatively affects customer experience, contributing to onboarding friction and potential customer attrition in an increasingly competitive digital banking environment (Validata, 2023).

Despite the existence of rule-based validation mechanisms, there remains limited localized research within Namibia exploring the application of predictive modelling techniques—particularly those incorporating behavioural data—to enhance data quality at the point of entry (Nshimba & Lutu, 2023). This study is therefore motivated by the need to develop a proactive, data-driven approach that improves onboarding

data integrity, enhances operational efficiency, and supports national initiatives aimed at strengthening secure and reliable digital financial services.

1.5 Significance of the Study

This research provides value to the Namibian financial ecosystem by introducing a proactive data governance and predictive data quality framework for customer onboarding processes. The study offers policymakers data-driven insights that may support the development of sector-wide standards and benchmarking practices, particularly as Namibian banks increasingly adopt digital and machine learning-based technologies.

Commercial banks will benefit from reduced operational costs through minimizing detection lag and decreasing reliance on manual data remediation. The proposed model serves as a technical safeguard that can improve onboarding efficiency, enhance data integrity, and strengthen long-term institutional competitiveness (Redman, 2022; Kaur & Singh, 2021).

The Bank of Namibia, as the national regulator, may benefit from improved data quality across the financial sector, as strengthened onboarding data integrity supports compliance with Anti-Money Laundering (AML) and Know Your Customer (KYC) requirements under the Financial Intelligence Act (2012). This also aligns with broader national objectives aimed at advancing a secure and resilient digital financial ecosystem (Bank of Namibia, 2023).

Retail customers are expected to benefit from a more seamless onboarding experience characterised by reduced processing delays and fewer application rejections. Real-time detection of data entry errors enables immediate correction during the onboarding process, thereby improving user experience and reducing onboarding friction (Validata, 2023).

The academic community benefits from this study through its contribution to the limited body of knowledge on behavioural data analytics and machine learning

applications in banking within the Southern African context. It provides a foundation for further research on predictive data quality management in multi-channel and digital-first environments (Nshimba & Lutu, 2023).

Fraud and compliance officers may also benefit from the integration of behavioural indicators such as typing patterns and input timing, which can enhance the identification of anomalous or high-risk data entry behaviour. This adds an additional layer of support for fraud detection mechanisms and strengthens overall system integrity (Monaco, 2020).

1.6 Limitations of the Study

This study acknowledges several limitations related to data access, geographical scope, and technical implementation.

Access to live banking data is restricted due to privacy regulations and the sensitive nature of financial information. As a result, the study will rely on anonymised historical datasets and simulated data to ensure compliance with regulatory and ethical requirements while still enabling model development and evaluation.

The research is geographically focused on commercial banks operating in Windhoek, which may limit the generalisability of findings to rural banking environments or other regulatory contexts. However, the study is intended to provide a foundational predictive model that can be adapted and extended in future research across different regions.

The study is designed as a prototype model and does not involve full-scale deployment within operational banking systems. Variations in institutional IT infrastructure and system architecture may therefore affect real-world implementation, although the study focuses primarily on feasibility and predictive performance.

Furthermore, the research is limited to the data ingestion phase of retail onboarding and does not extend to downstream processes such as back-office validation,

workflow automation, or long-term data governance. This focused scope enables a detailed analysis of data quality at the point of entry.

Finally, behavioural indicators such as typing speed, input latency, and dwell time may be influenced by external factors including device type, user environment, and network conditions. These factors will be considered during preprocessing and model evaluation to reduce potential bias in performance outcomes.

1.7 Delimitations of the Study

To ensure the study remains focused and methodologically manageable, several boundaries have been intentionally defined.

The study is limited to commercial banks operating within Namibia and excludes non-bank financial institutions such as micro-lenders, insurance companies, credit unions, and informal financial service providers. This ensures consistency in the regulatory environment and onboarding processes.

Geographically, the study is confined to urban Namibia, with a particular focus on Windhoek, where digital banking adoption and hybrid onboarding systems are most prevalent.

The research considers data entry behaviours across two onboarding channels, namely digital self-service platforms and assisted in-branch onboarding processes. It focuses exclusively on individual retail customers and excludes corporate entities, small and medium enterprises, and non-profit organisations due to differences in onboarding complexity and documentation requirements.

From a data perspective, the study is restricted to structured onboarding data and selected behavioural indicators associated with user interaction during data entry. It does not include biometric authentication, unstructured data formats, or document image processing.

Technically, the study focuses solely on predictive data quality assessment at the point of data entry. It does not extend to downstream processes such as back-office validation, data cleansing, workflow automation, or post-onboarding data lifecycle management.

Chapter 2 - LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature on data quality management in the banking sector, with a focus on customer onboarding processes and predictive analytics. It examines both academic and industry perspectives on data quality frameworks, machine learning applications, and digital onboarding practices.

Particular attention is given to developments in African banking environments and the gaps that exist within the Namibian context. The review also explores the shift from manual to digital onboarding and the growing role of machine learning in improving data accuracy and efficiency. In addition, it considers the emerging use of behavioural data, such as keystroke dynamics, as a means of assessing data reliability. The chapter concludes by identifying the research gap that this study aims to address.

2.2 Conceptual Framework

The conceptual framework for this study is grounded in the “Garbage In, Garbage Out” (GIGO) principle, which emphasizes the importance of maintaining high-quality data at the point of entry. The framework is designed to improve data integrity in the banking onboarding process by shifting from reactive error correction to proactive prevention during data ingestion.

A central concept in this study is that the most effective stage to ensure data quality is at the point of data entry. Identifying and addressing errors at this stage reduces delays, operational costs, and compliance risks associated with late error detection in traditional banking systems.

The framework also incorporates entry-behavioural analytics, which suggests that user interaction patterns during data entry can provide insights into the reliability of

submitted information. Factors such as typing patterns and input timing may indicate uncertainty, inconsistency, or unusual behaviour, allowing potential issues to be identified in real time.

To support this approach, the study draws on established data quality management standards, such as the International Monetary Fund Data Quality Assessment Framework as a foundation for defining data quality dimensions. Rather than relying on a single historic framework, this study adapts selected dimensions to suit a machine learning-based onboarding environment. The dimensions used include completeness, accuracy, consistency, and timeliness.The final reliability assessment is calculated using a composite data quality score:

$$\mathbf{Q = w_1C + w_2A + w_3S + w_4T}$$

Where Q represents the overall data quality score, C denotes completeness, A denotes accuracy, S denotes consistency, and T denotes timeliness. The weights (w_1–w_4) represent the relative importance assigned to each dimension within the model.

By integrating this score with entry-behavioural analytics, the framework evaluates not only the data being entered, but also the reliability of the data entry process itself. This enables the proposed model to identify potential data quality issues early and support improved decision-making during customer onboarding.

2.3 Review of Related Studies

Human-Centric Errors in Banking Systems

The literature establishes that “dirty data” is a significant operational and financial burden in banking systems, extending beyond simple technical errors to represent a core risk in financial operations. Research distinguishes between institutional errors, such as transposition mistakes made by staff during manual data entry, and customer-related errors, which typically involve incomplete fields or missing information during self-service onboarding.

These data quality issues have direct consequences on customer behaviour. For instance, Validata (2023) reports that poor data quality contributes to onboarding abandonment rates of between 20% and 30%, leading to reduced trust in banking

systems. More recent studies indicate that data trust issues remain a key barrier to the adoption of artificial intelligence in financial systems (Sichone, 2017).

Machine Learning in Data Quality

Traditional approaches to data quality have largely focused on reactive methods, where data is corrected after it has already been stored within systems. However, recent research indicates a clear shift toward proactive, real-time data quality assessment using machine learning techniques in order to prevent errors at the point of entry rather than correcting them post hoc (Kaur & Singh, 2021; Chen & Guestrin, 2016).

Studies in financial systems show that machine learning and data governance integration can significantly improve operational efficiency by reducing error rates and improving decision-making accuracy. Predictive analytics has also been associated with improvements in fraud detection and onboarding efficiency through early identification of anomalous or inconsistent data patterns (Lundberg & Lee, 2017; Redman, 2022). In addition, supervised learning models such as Random Forest and gradient boosting techniques have demonstrated high predictive performance in structured financial datasets, particularly in classification and risk detection tasks (Chen & Guestrin, 2016; Taylor & Francis, 2025).

Despite these advancements, implementation in developing banking environments such as Namibia remains limited due to challenges including fragmented data systems, limited integration of advanced analytics, and organisational reliance on rule-based validation approaches (Bank of Namibia, 2023; Nshimba & Lutu, 2023).

Keystroke Dynamics and Input Latency

The field of passive biometrics provides a non-intrusive approach to assessing data reliability by analysing human-computer interaction patterns during data entry. This includes behavioural indicators such as input latency, typing speed, and correction frequency, which can reflect user uncertainty or inconsistency.

Studies show that these behavioural features, when combined with advanced feature engineering techniques, can achieve strong predictive performance, with some models reaching up to 95% Area Under the Curve (AUC) in classification tasks (Mobilewalla). These techniques are increasingly used to reduce processing bottlenecks in financial systems (Proxy Group, 2024), although their effectiveness depends on balancing automation with user experience considerations (Wavetec, 2025).

In the Namibian context, emerging studies indicate that integrating behavioural analytics into financial models can improve efficiency and decision-making. These approaches are being explored within broader financial modelling frameworks such as credit risk assessment and non-performing loan prediction (Bank of Namibia, 2024; University of Namibia).

2.3.1 Research Gap

While existing literature demonstrates significant progress in the application of machine learning for fraud detection, credit scoring, and fintech adoption, limited research exists on its application to real-time data quality assessment during customer onboarding, particularly in the Namibian context. Current local studies primarily focus on credit risk and digital adoption, with little emphasis on data quality at the point of entry.

Specifically, there is a lack of models that assign a real-time reliability score to onboarding data in order to detect and prevent errors as they occur. Furthermore, limited research has examined data entry challenges within Namibia's regulatory environment, particularly under the Financial Intelligence Act (2012). There is also a gap in studies comparing data quality across different onboarding channels, such as in-branch staff entry and customer self-service platforms.

This study addresses these gaps by developing a predictive data quality scoring model tailored to the Namibian banking environment.

2.4 Summary

The reviewed literature confirms that while the global banking industry is rapidly digitising, it continues to rely heavily on reactive audit mechanisms and post-entry data cleaning processes. Although existing research demonstrates the effectiveness of machine learning in improving general data quality, there remains a clear gap in models specifically designed for the Namibian customer onboarding environment.

In particular, the use of machine learning for real-time assessment of human-centred data entry behaviour remains under-explored within the Southern African context. Current studies largely focus on fraud detection and credit scoring, with limited attention given to proactive data quality evaluation at the point of entry.

This study seeks to address this gap by developing a predictive data quality scoring model grounded in the Data Quality Assessment Framework (DQAF) and adapted to the Namibian banking context. The proposed approach shifts the focus from reactive data correction to proactive, real-time assessment of data reliability during customer onboarding.

Chapter 3 – METHODOLOGY

3.1 Introduction

This chapter outlines the systematic approach used to develop the Predictive Data Quality Scoring Model. The study adopts a quantitative research design under the Design Science Research (DSR) methodology, which is suitable for the development and evaluation of artefact-based solutions to real-world problems.

To guide the development of the machine learning model, the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework is applied as a structured model-building approach. CRISP-DM is used to guide the phases of business

understanding, data understanding, data preparation, modelling, and evaluation within the predictive modelling process.

The study uses Python programming tools, including scikit-learn and XGBoost, to implement machine learning techniques for model training, evaluation, and validation. By applying these methods to anonymised and simulated banking data, the study translates theoretical data quality dimensions into a measurable predictive scoring system.

3.2 Methodological Approach

This research is grounded in a positivist paradigm and adopts a quantitative research design under the Design Science Research (DSR) methodology. DSR is appropriate for this study as it focuses on the development of an artefact—in this case, a predictive data quality scoring model—to solve a practical problem in banking data quality management.

The development of the model follows the CRISP-DM framework, which structures the machine learning process into phases including business understanding, data preparation, modelling, and evaluation. In this study, CRISP-DM guides the technical implementation of the predictive model.

Behavioural indicators such as dwell time (duration a key is pressed), flight time (time between keystrokes), input latency, and correction frequency are used as measurable features of user interaction during data entry.

Model performance is evaluated using classification metrics. The primary benchmark for success is the Area Under the Curve (AUC-ROC), with a score of 0.90 or higher considered indicative of strong predictive performance.

3.3 Research Methods

3.3.1 Data Collection Methods

This study uses both primary and secondary data sources.

Primary data is collected through a simulated onboarding environment where participants act as either customers or bank staff. During data entry, a logging script captures behavioural metrics such as input latency, dwell time, flight time, and correction frequency. These variables represent real-time user interaction patterns during onboarding. A purposive sampling technique is used to select participants familiar with banking forms to ensure realistic data entry behaviour.

Secondary data consists of anonymised historical error logs obtained from banking-related datasets. These logs are used to identify common human-related data entry errors and support model training.

3.3.2 Model Performance Evaluation Metrics

The data analysis process begins with preprocessing, including data cleaning, encoding, imputation, and scaling to ensure a high-quality dataset for modelling. Feature engineering is applied to transform raw behavioural signals into meaningful indicators such as average typing cadence and a field-level uncertainty index.

Predictive modelling follows a supervised learning approach using an 80/20 train-test split with cross-validation to ensure robustness. Logistic Regression is used as a baseline model, while Random Forest and XGBoost are used to capture complex non-linear relationships between behavioural inputs and data quality outcomes.

Model performance is evaluated using a Confusion Matrix, Accuracy, Precision, Recall, and F1-Score to balance classification performance. The Area Under the Curve (AUC-ROC) is used as the primary evaluation metric.

To ensure interpretability, SHAP (SHapley Additive exPlanations) is applied to explain feature contributions to model predictions. In addition, a detection lag

reduction metric is used to measure improvements in error detection speed compared to traditional manual validation approaches.

3.4 Ethical Considerations

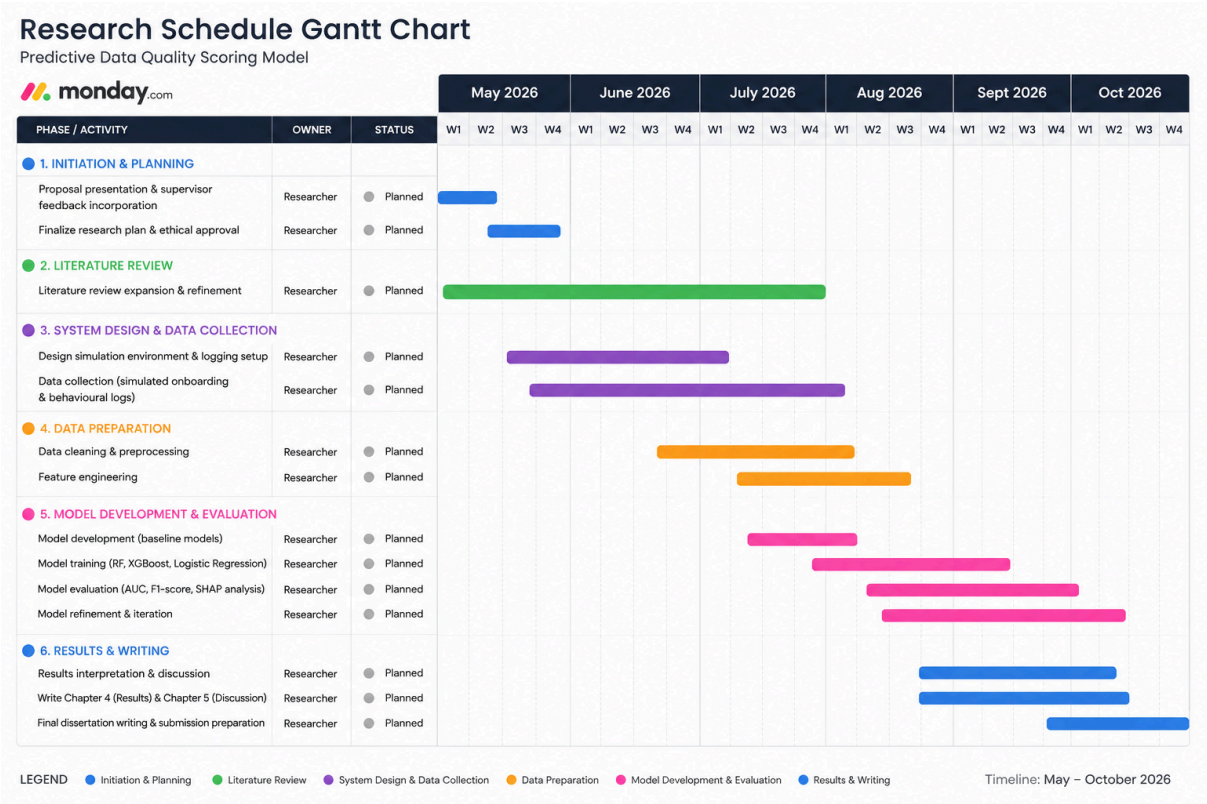
Informed Consent

Participants will be provided with a consent form outlining the purpose of the study, procedures, potential risks, and data handling practices in accordance with ethical research guidelines. Informed consent will be obtained from all participants prior to data collection, and participation will be entirely voluntary, with the option to withdraw at any stage.

Privacy and Confidentiality

All data collected during the study will be treated with strict confidentiality. Participant identities will be anonymised in all datasets, reports, and publications arising from the research. Access to the collected data will be restricted and securely stored to ensure compliance with data protection and ethical research standards.

3.4 Research Schedule



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